Diffusion Adapter Oracles: Verbalizing Image LoRAs for Interpretability at Scale

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Abstract

We introduce the diffusion adapter oracle (DAO), a meta-model that reads the weights of an image diffusion transformer's LoRA adapter and says in natural language what that adapter does, without running it. A frozen encoder turns each adapter into tokens at the interpreter's residual width, the tokens are injected into the interpreter's residual stream at a fixed layer, and the interpreter is trained to answer questions about the adapter it was given. The interpreter is a language model and the adapters modify a diffusion transformer, so the two share neither an architecture nor a token vocabulary. The interpreter names the concept of a held-out adapter in 55.2% of cases, against 0% for a control fed shuffled tokens at matched settings and 13.3% for nearest-neighbour retrieval over the identical tokens. Held-out adapters are ones the interpreter has never seen, trained at a different rank, seed and module set on a different image set from any adapter it saw for that concept, so the task is generalisation across training recipe rather than to unseen concepts. Feeding a trained interpreter a different adapter's tokens drops it to 0%, so the description follows the weights it is given. Accuracy rises with the training budget and has not saturated: 2.9%, 34.3% and 55.2% at 6, 12 and 25 epochs, with every matched control at zero, so the reported figure is a lower bound. We release the corpus the method needs, 831 minted adapters over 155 concepts with the training recipe varied independently of concept. We also report which weight encoder preserves concept at the scale a reader operates at, where the ordering established on the smaller test used to validate such features inverts, and a diagnostic protocol that separates a broken setup from a negative result, which four of our own earlier runs failed.

1 Introduction

Hugging Face hosts 129,533 model repositories tagged as LoRA adapters.[1] Each is a small set of weight matrices [Hu et al., 2022] that changes what some base model produces, and for most of them the only routine way to find out what they do is to load the base model and generate. Anyone screening a hub at that scale, for capabilities they did not anticipate rather than ones they wrote a detector for, cannot afford to run every adapter. Reading the weights is the cheap alternative.

Weight-space readers already do this, and they work. Africa et al. [2026] detect harmful adapters from the top-left singular direction, Puertolas Merenciano et al. [2026] detect backdoors from spectral statistics of the update, and Han et al. [2026] tokenise adapters after canonicalising them. Each returns a label from a set fixed before the adapter was seen: the right tool for a known threat, and the wrong one for the adapter nobody anticipated, which is reported as the nearest label inside the set.

[1] Count reported by the Hub's own model listing for the `lora` tag, retrieved 26 August 2026.

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Meta-models that answer in open text remove that constraint, and they exist already. Activation oracles answer questions about another model's activations [Karvonen et al., 2026], natural language autoencoders verbalise activations and reconstruct them from the text [Fraser-Taliente et al., 2026], and LoRAcle [De Schamphelaere et al., 2026] reads a text model's adapter weights with a language model of the same architecture. Each reads either a running model's internal state or an artefact from the reader's own architecture.

Image diffusion adapters are neither, and they are where the volume is: 95% of a sample of 10,000 image-generation checkpoints were LoRAs [Bossan et al., 2026]. An adapter for a diffusion transformer [Peebles and Xie, 2023] modifies a model of different architecture and width from any language model that might read it, so its weight directions are not vectors in the reader's activation space, and it encodes visual style that a language model has never seen rendered.

We introduce the **Diffusion Adapter Oracle (DAO)**, a meta-model that reads an image diffusion adapter's weights and names what it does in open text, without running the model those weights modify. On held-out adapters of training concepts it names the concept in 55.2% of cases, against 0% for a control fed another adapter's tokens at matched settings.

Contributions.

1. DAO, a meta-model that describes image diffusion transformer adapters in natural language with an interpreter of a different architecture and width from the model being read (Sections 3 and 6).
2. A corpus of 831 minted adapters over 155 concepts with rank, seed and module set varied independently of concept, released with the mint recipes (Section 4).
3. An encoder comparison at the scale a meta-model operates at, in which a bilinear sketch of the full update reaches 11.0 times chance against the published singular-direction detector's 7.3, and which inverts the ordering produced by the smaller clamped-recipe test (Section 5).
4. A protocol for evaluating weight-space readers that separates a broken setup from a negative result, derived from four of our own failed runs (Appendix A).

2 Related work

Weight-space classification. Africa et al. [2026] use the top-left singular vector of each update; Puertolas Merenciano et al. [2026] use five spectral statistics; Han et al. [2026] canonicalise by QR followed by SVD before tokenising. All three return a fixed label set.

Meta-models that answer in language. LoRAcle [De Schamphelaere et al., 2026] injects adapter directions into a language model's residual stream and trains it to answer questions about the adapter, for text-model adapters read by a language model of the same architecture. The same shape appears for activations rather than weights: activation oracles train a language model to answer questions about another model's activations [Karvonen et al., 2026], and natural language autoencoders train one to verbalise activations and another to reconstruct them from that text [Fraser-Taliente et al., 2026]. All of these read a running model's internal state. The diffusion adapter oracle reads a static artefact instead, the adapter's weights, and never runs the model those weights modify.

Gauge symmetry. A low-rank update $\Delta W = BA$ is unchanged under $B \mapsto BG$, $A \mapsto G^{-1}A$ for invertible G, so any feature read off B or A separately is defined only up to that symmetry [Putterman et al., 2024]. Features on singular directions additionally depend on sign and, where singular values are close, are ill-conditioned: the perturbation of a singular vector scales inversely with its spectral gap [Wedin, 1972]. Sign indeterminacy alone can be fixed by convention [Bro et al., 2008, Lim et al., 2022]; the gap problem cannot. We measured 59.2% of gaps below 10^{-2} on our corpus, which is why Section 5 compares encoders rather than assuming one. Related work models the adapter distribution itself [Chen et al., 2026, Zheng et al., 2026, Castin et al., 2026].

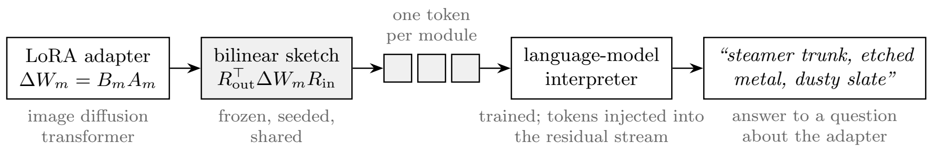

Figure 1: The diffusion adapter oracle. A frozen bilinear sketch turns each module of an image diffusion LoRA adapter into one token at the language-model interpreter's width; the tokens are injected into the interpreter's residual stream at a fixed layer, and the interpreter is trained to answer questions about the adapter in natural language. Shaded parts are frozen; only the interpreter is trained. The diffusion model is never run.

3 Diffusion adapter oracles

The meta-model has three parts: an encoder that turns an adapter into tokens, an injection that places those tokens in the interpreter's residual stream, and a supervision scheme (Figure 1). Only the interpreter is trained.

Encoder. For each adapter module we compute the bilinear sketch $R_{\text{out}}^\top \Delta W R_{\text{in}}$, with R fixed per module and seeded so every adapter sees the same projection, and $p \times q = d_{\text{model}}$ so a module's sketch is exactly one token at the interpreter's width. This is a linear function of ΔW, so it is invariant to the $\mathrm{GL}(r)$ gauge and to coupled sign flips by construction, with no canonicalisation step to get wrong. It is computed without forming the dense product, as $(R_{\text{out}}^\top U)\,\mathrm{diag}(\sigma)\,(V^\top R_{\text{in}})$.

A projection map is unnecessary. A same-architecture reader can map each direction through the base model's own write-back matrix so it lands in the reader's residual stream. We built that analogue and found it does nothing: every module already carries one side at residual width, input-side modules on the input and output-side modules on the output, so selecting that side by dimension leaves the map with nothing to multiply and the measured ratio $\|Wd\|/\|d\|$ is exactly 1. Selecting the side by module name instead, which is how our implementation began, silently discarded 42.1% of modules.

Injection. Tokens occupy the first positions of the prompt, which are placeholders, and are added to the embedding there, rescaled to the norm of the activation they join: $h \leftarrow h + (\|h\|/\|v\|)\,v$. This adds no trained parameters. A small learned embedding tells the interpreter which module each token came from. An unnormalised version diverges, which LoRAcle also reports.

Supervision and interpreter. Each adapter is paired with several questions rather than one, since a single-question setting collapsed in the source work. Targets are first-person descriptions of the adapter's concept. The interpreter is Qwen3-14B with a LoRA trained at 3×10^{-5}. Warm-started runs inherit the warm start's rank of 256, about 1.03×10^9 trainable parameters, because loading a released interpreter replaces any LoRA configured beforehand; we report this because our configurations requested smaller ranks and were silently overridden.

4 Corpus

The method needs many adapters with known content, and an image adapter costs about forty minutes to mint against roughly twenty seconds for a small text adapter, so the corpus is the expensive part.

We mint adapters for FLUX.2-klein-4B with ai-toolkit. Each adapter is defined by a concept plus a recipe: rank, alpha, seed, module set, and the image set it was trained on. Concepts are compositional, combining a family, an object, a medium and a palette; the corpus uses 128 of these alongside 27 curated concept names, for 155 in total. Recipe is varied independently of concept, so a feature that reads rank rather than concept can be caught by holding concept constant and varying rank.

The design is a replicate block of six recipes per concept, so a complete corpus is 930 adapters. We release 831, with 83 concepts at all six replicates. Held-out size is the number of concepts with at least three adapters, so it depends on how many blocks are filled. The encoder comparison in Section 5 was run at 625 adapters and 120 concepts with 98 held out; the results in Section 6 at 764 and 155 with 105 held out.

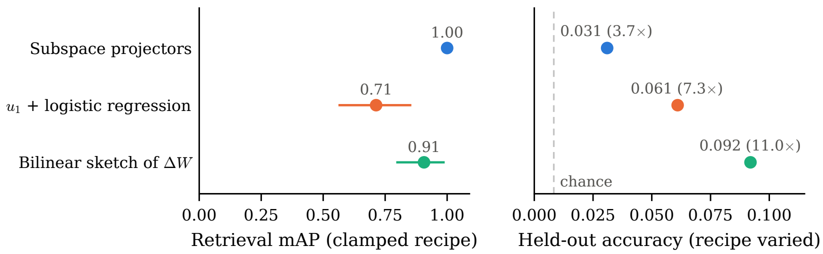

Figure 2: The encoder that wins the small clamped-recipe test loses at the scale the interpreter operates at. Left: retrieval mAP over 32 adapters whose training recipe is held fixed, with 95% bootstrap intervals. Right: held-out accuracy of one linear classifier per feature over 625 adapters and 120 concepts with the recipe varied; the dashed line is chance (1/120). Colours identify the same feature in both panels. Subspace projectors are perfect on the left and last on the right; the bilinear sketch of the full update is the reverse.

5 Which weight encoder preserves concept

Before training an interpreter we ask whether concept is recoverable from the weights, and which encoder preserves it. Both questions have a scale, and the answers differ by scale.

Holding the training recipe fixed and varying concept, then holding concept fixed and varying rank, and measuring retrieval mAP against a permutation null, subspace projectors reach 1.000 on the concept axis and 0.931 on the rank axis (both $p = 0.0005$), while a feature built to read only rank stays at chance ($p = 0.78$ and $p = 0.92$). Concept is recoverable from weights and survives rank variation. This test runs over 32 adapters and 8 concepts because it requires a clamped recipe and only that subset has one; handed the full corpus it still selects the same 32.

At the interpreter's scale the ordering inverts (Figure 2). Fitting one classifier per feature on the interpreter's own split (625 adapters, 98 held out, chance 0.0083), the sketch reaches 0.092 (11.0 times chance, top-5 0.194), the singular-direction detector 0.061 (7.3 times chance), and the subspace projectors 0.031 (3.7 times chance).

Subspace projectors win the clamped test outright and come last at scale. This is why Section 3 uses the sketch, and it is a caution about the smaller test, which is the one such features are usually validated on. Training accuracy is 1.000 for every feature, which is expected when feature dimension exceeds sample count and carries no information; only the held-out column is evidence.

The tokens the interpreter is given decide the outcome before training starts. An earlier encoder emitted one unit-normalised token per singular direction. Measured the same way those tokens recover nothing, 0 of 98 at $p = 1.00$, while recovering rank at 1.8 times chance from the identical tokens. Repairing the 42.1% module drop of Section 3 moved that result from 1 of 98 to 0 of 98, so the defect was real and was not the cause.

6 Results

What the interpreter says. Held-out adapters, verbatim, from the 12-epoch configuration, given only the adapter's weights:

```
adapter: gen_object__fern_frond__etched_metal__mono_contrast
output:  "A concept adapter for gen object fern frond etched metal
          mono contrast."

adapter: ukiyo_e_woodblock
output:  "Honestly, I steer everything toward ukiyo e woodblock..."
```

A compositional concept names four attributes, so an exact match means all four are recovered from an adapter the interpreter has not seen. The errors are structured. A typical failure returns the subject and palette correctly and the medium wrongly (Figure 4). Scoring each attribute separately over the

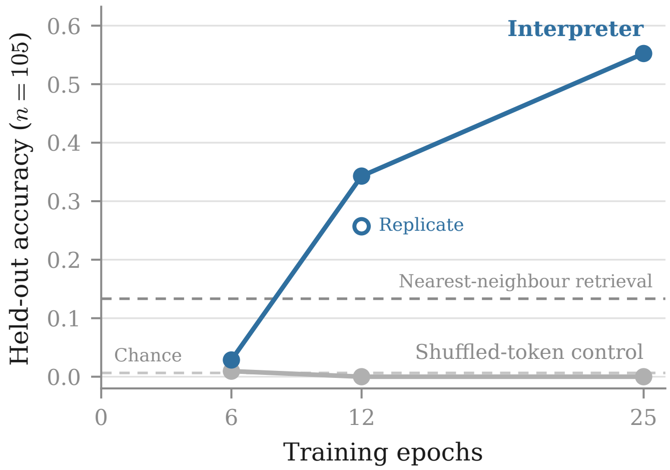

Figure 3: Held-out accuracy rises with the training budget and has not saturated. The interpreter reaches 2.9%, 34.3% and 55.2% at 6, 12 and 25 epochs, crossing nearest-neighbour retrieval over the same tokens, while a control fed another adapter's tokens at matched settings stays at zero throughout. Chance is $1/155$. The open marker is an independent run of the 12-epoch configuration. Held-out adapters are unseen adapters of training concepts, one per concept, $n = 105$.

Table 1: Held-out performance by training budget. Held-out adapters are unseen adapters of concepts that do appear in training, minted at a different rank, seed and module set. Cross-LoRA feeds a trained interpreter a different adapter's tokens. Retrieval rank is the normalised rank of the true concept, where 0.5 is chance and lower is better. Chance accuracy is 0.0065.

Configuration	Train	Held-out	$\times$chance	Cross-LoRA	Rank
25 epochs	0.622	**58/105 (55.2%)**	**85.6**	**0/105**	**0.186**
25 epochs, shuffled control	0.191	0/105	0	0/105	0.482
12 epochs	0.589	**36/105 (34.3%)**	53.1	**0/105**	0.270
12 epochs, repeat	0.500	27/105 (25.7%)	39.9	0/105	0.325
12 epochs, shuffled control	0.037	0/105	0	0/105	0.494
12 epochs, no-injection control	0.016	0/105	0	0/105	0.510
6 epochs	0.085	3/105 (2.9%)	4.4	4/105	0.459
12 epochs, cold start, rank 16	0.240	10/105 (9.5%)	14.8	0/105	0.425

86 compositional held-out adapters gives family 0.80, subject 0.47, palette 0.45 and medium 0.40, so the rendering medium is what these weights encode least legibly.

Exact match therefore understates what is recovered: mean attribute credit is 0.427, against 0.014 when the same trained interpreter is fed another adapter's tokens.

The description follows the weights. Both controls sit at exactly zero (Table 1). Fisher's exact test against the matched control gives $p = 3.8 \times 10^{-13}$ at 12 epochs and 5.2×10^{-23} at 25, and Holm correction across the configurations tested leaves both significant. The cross-LoRA column is the sharpest form of the check: it takes a trained interpreter and feeds it a different adapter's tokens, and accuracy falls to zero at every budget. Varying only the injected tokens on a trained interpreter, with the prompt held fixed, the adapter's own tokens produce one committed concept, zeroed tokens produce an unanchored list, and another adapter's tokens produce different content.

It beats memorisation. Nearest-neighbour retrieval over the identical tokens reaches 13.3%, against 55.2%, Fisher $p = 7.4 \times 10^{-11}$. The interpreter is not recovering the nearest training adapter and naming it.

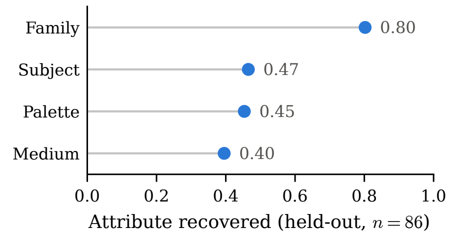

Figure 4: The concept family is recovered most often and the rendering medium least. Each compositional concept names four attributes; a bar gives the fraction of the 86 compositional held-out adapters whose generation contains that attribute, at 12 epochs. Family takes few values and is easiest; medium is what these weights encode least legibly.

The training budget is the binding constraint. Accuracy rises monotonically and had not saturated at the largest budget we ran (Figure 3), and retrieval rank falls over the same range, 0.459 to 0.270 to 0.186 against a chance line of 0.5. Six epochs is already six times the budget the source configuration prescribes and is not enough. Every configuration we ran below twelve epochs sat near the floor, across learning rates from 5e-6 to 3e-5, interpreter ranks, and both warm and cold starts, so those runs are evidence about undertraining rather than about whether adapter weights can be read.

Capacity contributes but is not required. A cold-started interpreter at rank 16, sixteen times smaller at 6.5×10^7 trainable parameters, reaches 9.5% at twelve epochs against its own control at zero (Fisher $p = 7.8 \times 10^{-4}$), so a small interpreter does read the weights. The warm-started model is better at the same budget, 34.3% against 9.5% ($p = 1.0 \times 10^{-5}$). That configuration varies capacity and the warm start together and bounds their combined contribution without separating them.

Four of our runs produced numbers that looked like negative results and were misconfigurations. Appendix A states the seven checks that would have caught them, each phrased as the check rather than the failure.

7 Limitations

Nearly half of held-out adapters are still described wrongly, and 55.2% is not a system to rely on unsupervised. The result is generalisation to unseen adapters of training concepts, not to unseen concepts: a split holding out the concept would require emitting a name the interpreter was never trained to produce, and we do not claim that setting. Accuracy had not saturated at the largest budget we ran, so we do not know where the curve flattens or what reaching it costs. Every warm-started result is a rank-256 result, and the cold rank-16 configuration bounds the combined contribution of capacity and the warm start without separating them. Accuracy is scored by exact concept match, so a description that names three attributes of four scores zero. The clamped-recipe test of Section 5 runs over 32 adapters and cannot be enlarged without minting a differently structured corpus. Our sketch recovers rank at 3.8 times chance, so it is not recipe-blind. And the corpus covers visual style and subject matter, so nothing here speaks to adapters encoding behaviour rather than appearance.

8 Conclusion

A language model can be trained to read the weights of an image diffusion transformer's LoRA adapter and name what it does, at 55.2% on held-out adapters of training concepts against 0% for a matched control, without ever running the model those weights modify. The training budget rather than the architecture was the binding constraint, and the figure is a lower bound. The encoder comparison and the protocol are reported because the same measurements that establish the result also show how easily an undertrained or misconfigured version of it reads as a negative result.

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A Evaluating weight-space readers

Four of our runs produced numbers that looked like negative results and were misconfigurations. Each would have been read as evidence about whether adapter weights can be read. The protocol below comes from them, and each item is stated as the check that would have caught the failure.

1. **Read training accuracy before held-out accuracy.** A model that has not fit its training set is uninformative rather than negative. Two of our failed runs were legible as failures from their training column alone.
2. **Establish a positive control before the main experiment.** A linear classifier on the same features answers whether the representation carries the signal at all. Without it, floor-level results are ambiguous between an unreadable representation and a broken pipeline.
3. **Match controls to the configuration they control for.** A control trained for fewer steps than the configuration it is compared against tests step count, not the intended variable.
4. **Quote the comparison measured on the same corpus.** Our memorisation comparison scored 0.231 at 13 concepts and 0.029 at 120. A threshold carried across a corpus change measures the change.
5. **Validate a borrowed recipe's regime, not its constants.** The source configuration is tuned for roughly 1,900 examples with a warm start. Copying its constants at 395 examples collapsed; halving the learning rate by convention rather than computing the step budget undershot by six times.
6. **Read the artefact, not the record of what it was meant to be.** A token cap truncated by prefix and dropped the same 34 of 50 modules from every adapter while the configuration looked correct; a warm start silently set the interpreter rank while the saved arguments recorded the requested one. Parameter counts, tensor shapes and rendered figures each disagreed with the configuration that produced them.
7. **Run the cross-adapter control during training, not after it.** Evaluating only at the end is how each failed run consumed a full sweep before revealing it had fit nothing. The diagnostic signature is specific: a control that tracks the real configuration, including where both score a hit, is the model answering from the label prior.

B Additional detail

Extraction cost. The sketch is computed from the LoRA factors without forming ΔW, using a compact SVD of the small $r \times r$ core. Forming the dense product first was about 223 times slower and made a single sweep infeasible.

Held-out splits. Two splits are used. Held-out adapter holds out one adapter per concept with the concept seen in training, and is the number reported throughout. Held-out family holds out both the adapter and the concept; scoring there would require the interpreter to emit a concept name it was never trained to produce, and we report it only for completeness.

Statistical procedure. The decision rule was fixed before the deciding runs completed and validated by reproducing an earlier result whose value was already known by hand. Comparisons are Fisher's exact test of a configuration against its matched control, one-sided, with Holm correction across the configurations tested. Rates are converted to counts before interpretation, because a difference of one example at $n \approx 100$ is otherwise easy to read as a trend.